Abstract: Active Skillference as Provenance-Bound Curriculum Infrastructure

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Active Inference and the Free Energy Principle (FEP) provide model-based accounts of belief updating, learning, and action under uncertainty. We present Active Skillference, a provenance-bound curriculum-generation and SkillTree-export system for teaching those formal ideas. The paper evaluates structural validity, quantitative provenance, citation-role coverage, and artifact reproducibility; it does not evaluate learner outcomes, establish a new theory of Active Inference, or present an intelligent tutoring system.

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The curriculum is expressed as code: a typed, validated directed acyclic graph of 630 skills across 111 subjects spanning all 8 strata (mathematics -> probability -> information theory -> variational methods -> the FEP -> active inference -> computation -> applications), connected by 1199 prerequisite edges with a maximum dependency depth of 75 (of which the substantive concept chain accounts for 33; the remaining depth is per-stratum review and mastery sequencing rather than conceptual prerequisite, as the methodology details). Its defining feature is content-provenance binding: every quantitative value shown to a learner is produced by a tested computational kernel and inserted through a typed claim token, never hand-typed, and the build refuses to export if a claim is unbacked or if a bare result number appears in learner prose, manuscript prose, or

correct numeric quiz answers. The contribution is therefore a systems and curriculum-infrastructure artifact: it makes a formal subject inspectable and deliverable, but does not claim that the resulting path is optimal for every learner.

The validated graph exports directly into SkillTree's data model (Project -> Subjects -> Skills with learning-path dependencies and quiz-gated completion), includes a scripted REST seeding path for a configured instance, and is mirrored by a local dashboard that exposes generated artifacts, figures, claim ledgers, scholarship audits, and graph diagnostics without taking ownership of learner progress or scoring from SkillTree. The result is a curriculum with a validator-backed artifact chain: re-running the kernels regenerates the claim ledger, figures, manuscript variables, SkillTree export, and learner-facing numbers, so the platform's teaching

claims remain bounded by what the code, citations, validators, and documented limitations actually support.